System Studies

ARPA-E DC Grids

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AMERICAN TERAWATT

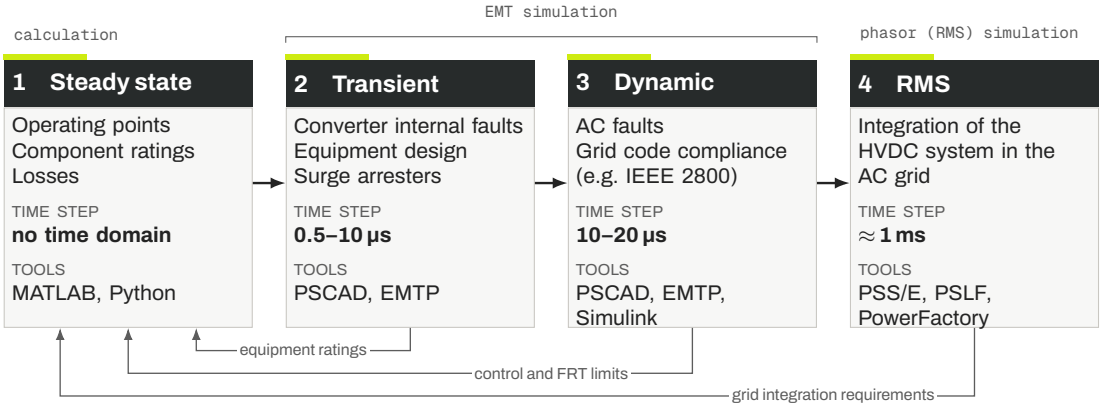
Abbreviations

AC	Alternating current
DC	Direct current
EMT	Electromagnetic transient
EMTP	Electromagnetic Transients Program
FRT	Fault ride-through
HVDC	High-voltage direct current
IEEE	Institute of Electrical and Electronics Engineers
PSCAD	Power Systems Computer Aided Design
PSLF	Positive Sequence Load Flow
PSS/E	Power System Simulator for Engineering
RMS	Root mean square

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Design process
System studies

Results of each study feed back into the steady-state design.



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Steady state
Operating points, ratings and losses

Calculation, no simulation tool

Algebraic calculation of the converter station at its steady-state operating points.

- Operating points across the power, AC voltage and DC voltage range
- Steady-state rating of the components
- Losses per component and for the station
- Starting point for all following studies

Tools	MATLAB, Python
Time step	no time domain
Results	Operating points, component ratings, losses

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Transient simulation
Converter internal faults

EMT simulation, detailed model

Transient behaviour of the system during converter internal faults.

- Converter internal faults, e.g. valve and converter bus faults
- Design of the equipment for the transient stresses
- Design of protective equipment such as surge arresters
- Returns updated ratings to the steady-state design

Tools	PSCAD, EMTP
Time step	0.5–10 µs
Results	Equipment stresses, arrester ratings

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Dynamic simulation
AC faults and grid code compliance

EMT simulation, control-oriented model

Dynamic behaviour of the system during AC faults.

- AC faults at and near the point of connection
- Grid code compliance defined by e.g. IEEE 2800
- FRT performance and control response
- Returns control and FRT limits to the design

Tools	PSCAD, EMTP, Simulink
Time step	10–20 µs
Results	Control performance, FRT behaviour, compliance evidence

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RMS model
Integration in the AC grid

Phasor simulation, grid-level model

Simulation of the integration of the HVDC system in the AC grid.

- HVDC system as part of the wider AC network
- Model handed over to the grid operator for planning studies
- Returns grid integration requirements to the design

Tools	PSS/E, PSLF, PowerFactory
Time step	≈ 1 ms
Results	RMS model for grid studies

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